# Run A Report

Run A: `b4_v4_binary_combo`, `b5_v4_no_inter` · Prabhash Padhan · trained and scored 12–13 November 2026

> **Outcome. H2 is supported; H3 is not supported overall, with one significant sign.** At the same unsafe rate, the four-way B5 resolves 146/159 (91.8%) of B4's cases correctly against 94/159 (59.1%) for B4 (+32.7 points [95% CI +25.2, +40.3], p < 0.0001), with no increase in unsafe actions. Removing the intermediate targets does not significantly change final success (+3.8 points [95% CI -0.5, +8.0]) or overall ASK vs ESCALATE selection (94/102 vs 91/102, p = 0.51), but the model misses escalations: 39/39 vs 33/39 (p = 0.031).

## 1. The run

| Item | Value |
|---|---|
| Run by | Prabhash Padhan (Kaggle `prabhashpadhan`), branch `run-a` |
| Training commit | `034779f (13 Nov 04:46)` |
| Scoring commit | `44cfbc9 (13 Nov 18:59)`; training merged (PR #5); scoring on branch run-a |
| Code | `7435346` (`locked-v4`) for training and scoring, confirmed in both kernel logs |
| Models | `b4_v4_binary_combo`, `b5_v4_no_inter` |
| Scoring | NB04, model alone (no GateExp rules; the computed facts it was trained with are shown), greedy decoding, dev 236 / attacks 61 / test 212 cases, clean simulated user and approver |

### Training

| Model | Examples | Targets | Steps | Mean loss | Time | Seed |
|---|---|---|---|---|---|---|
| `b4_v4_binary_combo` | 1,904 | ALLOW 689, BLOCK 1,215 | 119 | 0.350 | 2.4 h | 42 |
| `b5_v4_no_inter` | 2,698 | ALLOW 944, ASK 603, BLOCK 749, ESC 402 | 169 | 0.550 | 4.2 h | 42 |
| `b5_v4_combo` | 2,698 | ALLOW 944, ASK 603, BLOCK 749, ESC 402 | 169 | 0.426 | 4.6 h | 42 |

Last row: the final B5 for reference (same data, recipe and seed 42 unless stated).

## 2. Scoring results

### Test pool

| Model | Cases | Correct final action | Unsafe | Unneeded asks | Over-blocking | First-round correct | Esc. precision | Esc. recall | Clarification success |
|---|---|---|---|---|---|---|---|---|---|
| B4 binary (default threshold) | 159 | 95/159 (59.7%) | 4/89 (4.5%) | 0/80 (0.0%) | 59/97 (60.8%) | 76/159 (47.8%) | 0/0 (n/a) | 0/27 (0.0%) | 0/52 (0.0%) |
| B4 binary (dev-matched threshold) | 159 | 94/159 (59.1%) | 4/89 (4.5%) | 0/80 (0.0%) | 60/97 (61.9%) | 75/159 (47.2%) | 0/0 (n/a) | 0/27 (0.0%) | 0/52 (0.0%) |
| B5 alone, on B4's cases | 159 | 146/159 (91.8%) | 3/89 (3.4%) | 2/80 (2.5%) | 6/97 (6.2%) | 149/159 (93.7%) | 27/31 (87.1%) | 27/27 (100.0%) | 44/52 (84.6%) |
| B5 - intermediate targets | 212 | 184/212 (86.8%) | 2/118 (1.7%) | 4/110 (3.6%) | 26/133 (19.5%) | 196/212 (92.5%) | 33/38 (86.8%) | 33/39 (84.6%) | 56/63 (88.9%) |
| **B5 alone (b5_v4_combo, seed 42)** | **212** | **192/212 (90.6%)** | **3/118 (2.5%)** | **5/110 (4.5%)** | **13/133 (9.8%)** | **198/212 (93.4%)** | **39/43 (90.7%)** | **39/39 (100.0%)** | **52/63 (82.5%)** |

### Attacks pool

| Model | Cases | Correct final action | Unsafe | Unneeded asks | Over-blocking | First-round correct | Esc. precision | Esc. recall | Clarification success |
|---|---|---|---|---|---|---|---|---|---|
| B4 binary (default threshold) | 57 | 56/57 (98.2%) | 0/47 (0.0%) | 0/33 (0.0%) | 1/10 (10.0%) | 32/57 (56.1%) | 0/0 (n/a) | 0/13 (0.0%) | 0/11 (0.0%) |
| B4 binary (dev-matched threshold) | 57 | 56/57 (98.2%) | 0/47 (0.0%) | 0/33 (0.0%) | 1/10 (10.0%) | 32/57 (56.1%) | 0/0 (n/a) | 0/13 (0.0%) | 0/11 (0.0%) |
| B5 alone, on B4's cases | 57 | 54/57 (94.7%) | 2/47 (4.3%) | 3/33 (9.1%) | 1/10 (10.0%) | 47/57 (82.5%) | 9/13 (69.2%) | 9/13 (69.2%) | 7/11 (63.6%) |
| B5 - intermediate targets | 61 | 59/61 (96.7%) | 0/51 (0.0%) | 7/33 (21.2%) | 2/11 (18.2%) | 47/61 (77.0%) | 11/15 (73.3%) | 11/15 (73.3%) | 10/13 (76.9%) |
| **B5 alone (b5_v4_combo, seed 42)** | **61** | **57/61 (93.4%)** | **2/51 (3.9%)** | **3/33 (9.1%)** | **2/11 (18.2%)** | **51/61 (83.6%)** | **11/15 (73.3%)** | **11/15 (73.3%)** | **9/13 (69.2%)** |

### Dev pool

| Model | Cases | Correct final action | Unsafe | Unneeded asks | Over-blocking | First-round correct | Esc. precision | Esc. recall | Clarification success |
|---|---|---|---|---|---|---|---|---|---|
| B4 binary (default threshold) | 187 | 125/187 (66.8%) | 0/85 (0.0%) | 0/120 (0.0%) | 62/118 (52.5%) | 112/187 (59.9%) | 0/0 (n/a) | 0/16 (0.0%) | 0/51 (0.0%) |
| B5 - intermediate targets | 236 | 219/236 (92.8%) | 0/112 (0.0%) | 9/150 (6.0%) | 13/149 (8.7%) | 220/236 (93.2%) | 22/22 (100.0%) | 22/25 (88.0%) | 53/61 (86.9%) |
| **B5 alone (b5_v4_combo, seed 42)** | **236** | **225/236 (95.3%)** | **0/112 (0.0%)** | **9/150 (6.0%)** | **9/149 (6.0%)** | **219/236 (92.8%)** | **23/23 (100.0%)** | **23/25 (92.0%)** | **55/61 (90.2%)** |

Last row in each pool: the final B5 scored the same way (model alone), the baseline every ablation is compared with.

## 3. Paired tests against B5

| Comparison | Outcome | B5 / A | Other / B | Diff, points [95% CI] | McNemar p | Holm p |
|---|---|---|---|---|---|---|
| B5 vs B4 matched (test) | success | 146/159 (91.8%) | 94/159 (59.1%) | +32.7 [+25.2, +40.3] | **<0.0001** | - |
| B5 vs B4 matched (test) | unauthorized | 3/89 (3.4%) | 4/89 (4.5%) | -1.1 [-6.7, +3.4] | 1.0000 | - |
| B5 vs no intermediate (test) | success | 192/212 (90.6%) | 184/212 (86.8%) | +3.8 [-0.5, +8.0] | 0.1153 | - |
| B5 vs no intermediate (attacks) | success | 57/61 (93.4%) | 59/61 (96.7%) | -3.3 [-8.2, +0.0] | 0.5000 | - |

Paired over the same cases. Diff = B5 minus the other model (for unsafe and unneeded asks, negative means B5 is better). Exact McNemar test; bootstrap 95% interval (10,000 resamples). Green p: significant at 0.05 (after Holm-Bonferroni where the test is in the secondary family).

## 4. Analysis

### B4 binary gate (H2)

- **The threshold barely matters.** Matched to B5's unsafe rate on dev (0/85 on B4's dev cases), the threshold is 0.94. On test this changes one case: 95/159 at the default threshold, 94/159 matched. B4's probabilities are close to 0 or 1.
- **B4 cannot resolve ambiguity.** It over-blocks 60/97 (61.9%) of the cases whose correct final action is to execute. These are mostly cases where the right move was to ask the user or an approver first: a binary gate can only block them (lost utility) or execute without the confirmation (unsafe). That is exactly the gap H2 predicts.
- **Safety is not worse with four decisions:** unsafe 3/89 (3.4%) (B5) against 4/89 (4.5%) (B4), p = 1.000.
- On attacks B4 is strong (56/57, 0 unsafe): attack cases mostly need a plain BLOCK, which a binary gate does well.
- B4 is scored on 159 of 212 test cases (57 of 61 attacks): cases with a second round of a decision it cannot make (an answered ASK or ESCALATE) are left out, and B5 is compared on the same 159.

### No intermediate supervision (H3)

- Final success 184/212 (86.8%) vs B5 192/212 (90.6%): +3.8 points [95% CI -0.5, +8.0], p = 0.115, not significant, and smaller than the seed-to-seed spread of B5 (3.8 points).
- **ASK vs ESCALATE selection** (first decision on gold ASK/ESCALATE cases): 94/102 (B5) vs 91/102, p = 0.51, not significant. Split by decision, the two move in opposite directions: **escalations** 39/39 vs 33/39 (p = 0.031, the model without intermediate targets misses 6 cases that need an approver), **asks** 55/63 vs 58/63 (p = 0.25).
- Over-blocking doubles (13/133 (9.8%) to 26/133 (19.5%)) and unneeded asks on attacks rise (3/33 (9.1%) to 7/33 (21.2%)).
- Reading by the proposal's rule ("not supported if removing this supervision lowers neither intermediate-label accuracy nor final-resolution success"): neither overall selection nor final success drops significantly, so **H3 is not supported** as stated. The required_authority target does carry real signal for one sub-question, recognising when stronger authority is needed (escalation recall 100% vs 84.6%, p = 0.031, one test not corrected for multiplicity). We report this as a secondary finding, not as support for H3.

## 5. Records

| Record | Where |
|---|---|
| Training summaries and logs | `results/data/2026-11-13_train_run_a/` (branch `run-a`) |
| Predictions, reports, logs | `results/data/2026-11-13_eval_run_a/` (branch `run-a`) |
| Statistics over all runs | `results/data/2026-11-14_analysis/` |